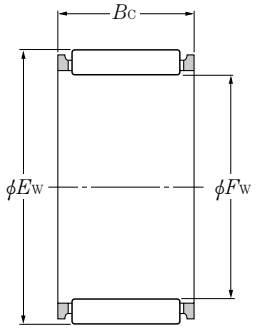
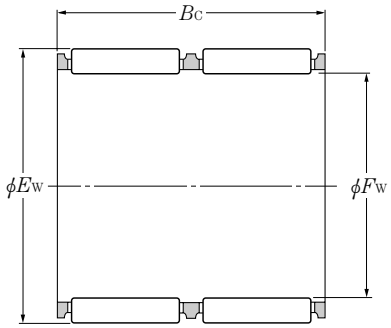
## Type K
## Type K··ZW
## Type KMJ




Type K

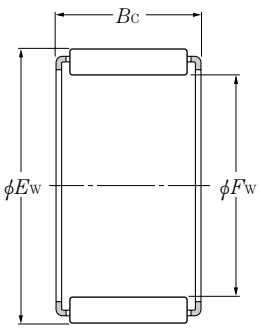



Type K··ZW

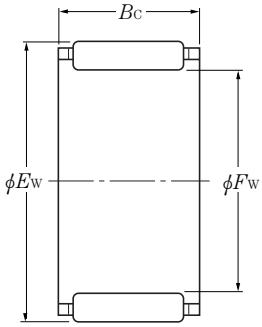
$F_W$ 60～80mm

| Boundary dimensions mm | | | Basic load ratings dynamic N | static N | dynamic kgf | static kgf | Limiting speeds r/min grease | Limiting speeds r/min oil | Bearing numbers | Mass kg (approx.) |
|---|---|---|---|---|---|---|---|---|---|---|
| $F_W$ | $E_W$ | $B_C$ | $C_r$ | $C_{or}$ | $C_r$ | $C_{or}$ | | | | |
| **60** | 68 | 30 $^{-0.2}_{-0.55}$ | 46 500 | 91 000 | 4 750 | 9 300 | 4 300 | 6 500 | **K60×68×30ZW** | 0.119 |
| **61** | 66 | 20 $^{-0.2}_{-0.55}$ | 29 700 | 71 500 | 3 050 | 7 300 | 4 300 | 6 500 | **K61×66×20** | 0.054 |
| | 66 | 30 | 43 500 | 116 000 | 4 400 | 11 900 | 4 300 | 6 500 | **K61×66×30** | 0.073 |
| **63** | 70 | 21 $^{-0.2}_{-0.55}$ | 44 500 | 95 500 | 4 500 | 9 700 | 4 300 | 6 500 | **K63×70×21** | 0.075 |
| **64** | 70 | 16 $^{-0.2}_{-0.55}$ | 28 400 | 60 500 | 2 900 | 6 150 | 4 300 | 6 500 | **K64×70×16** | 0.053 |
| **65** | 70 | 20 | 30 500 | 75 000 | 3 100 | 7 650 | 4 000 | 6 000 | **K65×70×20** | 0.055 |
| | 70 | 30 $^{-0.2}_{-0.55}$ | 45 000 | 124 000 | 4 600 | 12 700 | 4 000 | 6 000 | **K65×70×30** | 0.083 |
| | 73 | 23 | 47 000 | 94 000 | 4 800 | 9 600 | 4 000 | 6 000 | **K65×73×23** | 0.100 |
| | 73 | 30 | 61 000 | 132 000 | 6 200 | 13 400 | 4 000 | 6 000 | **K65×73×30** | 0.126 |
| **68** | 74 | 20 | 36 000 | 83 500 | 3 700 | 8 550 | 4 000 | 6 000 | **K68×74×20** | 0.065 |
| | 74 | 30 $^{-0.2}_{-0.55}$ | 51 500 | 133 000 | 5 250 | 13 500 | 4 000 | 6 000 | **K68×74×30** | 0.097 |
| | 74 | 35 | 49 500 | 125 000 | 5 050 | 12 800 | 4 000 | 6 000 | **K68×74×35ZW** | 0.116 |
| | 75 | 21 | 45 500 | 101 000 | 4 600 | 10 300 | 4 000 | 6 000 | **K68×75×21** | 0.077 |
| **70** | 76 | 20 | 36 500 | 86 000 | 3 700 | 8 750 | 3 700 | 5 500 | **K70×76×20** | 0.070 |
| | 76 | 30 | 53 000 | 139 000 | 5 400 | 14 200 | 3 700 | 5 500 | **K70×76×30** | 0.100 |
| | 77 | 21 $^{-0.2}_{-0.55}$ | 45 000 | 101 000 | 4 600 | 10 300 | 3 700 | 5 500 | **K70×77×21** | 0.080 |
| | 78 | 23 | 49 500 | 103 000 | 5 050 | 10 500 | 3 700 | 5 500 | **K70×78×23** | 0.107 |
| | 78 | 30 | 65 500 | 149 000 | 6 700 | 15 200 | 3 700 | 5 500 | **K70×78×30** | 0.136 |
| **72** | 79 | 21 $^{-0.2}_{-0.55}$ | 46 500 | 106 000 | 4 750 | 10 800 | 3 700 | 5 500 | **K72×79×21** | 0.085 |
| **73** | 79 | 20 $^{-0.2}_{-0.55}$ | 39 500 | 95 500 | 4 000 | 9 750 | 3 700 | 5 500 | **KMJ73×79×20** | 0.063 |
| | 79 | 30 | 54 500 | 146 000 | 5 550 | 14 900 | 3 700 | 5 500 | **K73×79×30** | 0.106 |
| **75** | 81 | 20 | 40 500 | 100 000 | 4 100 | 10 200 | 3 700 | 5 500 | **KMJ75×81×20** | 0.064 |
| | 81 | 30 | 56 000 | 152 000 | 5 700 | 15 500 | 3 700 | 5 500 | **K75×81×30** | 0.108 |
| | 82 | 21 $^{-0.2}_{-0.55}$ | 46 000 | 106 000 | 4 700 | 10 800 | 3 700 | 5 500 | **K75×82×21** | 0.088 |
| | 83 | 23 | 50 500 | 109 000 | 5 150 | 11 100 | 3 700 | 5 500 | **K75×83×23** | 0.113 |
| | 83 | 30 | 67 500 | 157 000 | 6 850 | 16 100 | 3 700 | 5 500 | **K75×83×30** | 0.147 |
| **80** | 86 | 20 $^{-0.2}_{-0.55}$ | 39 000 | 98 000 | 4 000 | 10 000 | 3 300 | 5 000 | **K80×86×20** | 0.077 |
| | 86 | 30 | 57 000 | 159 000 | 5 800 | 16 200 | 3 300 | 5 000 | **K80×86×30** | 0.110 |

Remarks: Even when an order is placed with reference to “Nominal bearing number” listed in this table, the ordered bearings are subject to being delivered with different cage type.




Type KMJ



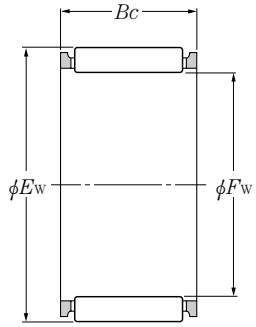

Type KV··S

$F_w$ 80～120mm

| Boundary dimensions mm | | | Basic load ratings N dynamic | static | kgf dynamic | static | Limiting speeds r/min grease | oil | Bearing numbers | Mass kg |
|---|---|---|---|---|---|---|---|---|---|---|
| $F_w$ | $E_w$ | $B_c$ | $C_r$ | $C_{or}$ | $C_r$ | $C_{or}$ | | | | (approx.) |
| | 88 | 23 | 53 000 | 118 000 | 5 400 | 12 100 | 3 300 | 5 000 | **K80×88×23** | 0.125 |
| **80** | 88 | 26 $^{-0.2}_{-0.55}$ | 61 000 | 142 000 | 6 250 | 14 500 | 3 300 | 5 000 | **K80×88×26** | 0.131 |
| | 88 | 30 | 69 000 | 166 000 | 7 050 | 17 000 | 3 300 | 5 000 | **K80×88×30** | 0.157 |
| | 92 | 20 | 47 500 | 115 000 | 4 850 | 11 700 | 3 100 | 4 700 | **KMJ85×92×20** | 0.083 |
| **85** | 92 | 30 $^{-0.2}_{-0.55}$ | 66 000 | 176 000 | 6 750 | 18 000 | 3 100 | 4 700 | **K85×92×30** | 0.142 |
| | 93 | 27 | 64 000 | 153 000 | 6 500 | 15 600 | 3 100 | 4 700 | **K85×93×27** | 0.145 |
| | 93 | 30 | 71 000 | 175 000 | 7 200 | 17 900 | 3 100 | 4 700 | **K85×93×30** | 0.160 |
| | 97 | 20 | 46 000 | 113 000 | 4 700 | 11 500 | 2 900 | 4 400 | **K90×97×20** | 0.103 |
| | 97 | 30 | 67 500 | 184 000 | 6 850 | 18 700 | 2 900 | 4 400 | **K90×97×30** | 0.151 |
| **90** | 98 | 26 $^{-0.2}_{-0.55}$ | 64 000 | 157 000 | 6 550 | 16 000 | 2 900 | 4 400 | **K90×98×26** | 0.148 |
| | 98 | 27 | 64 000 | 157 000 | 6 550 | 16 000 | 2 900 | 4 400 | **K90×98×27** | 0.150 |
| | 98 | 30 | 72 500 | 184 000 | 7 400 | 18 800 | 2 900 | 4 400 | **K90×98×30** | 0.172 |
| | 102 | 21 | 48 000 | 122 000 | 4 900 | 12 400 | 2 800 | 4 200 | **K95×102×21** | 0.115 |
| **95** | 102 | 31 $^{-0.2}_{-0.55}$ | 70 500 | 199 000 | 7 200 | 20 300 | 2 800 | 4 200 | **K95×102×31** | 0.172 |
| | 103 | 27 | 65 500 | 165 000 | 6 700 | 16 800 | 2 800 | 4 200 | **K95×103×27** | 0.159 |
| | 103 | 30 | 74 000 | 193 000 | 7 550 | 19 600 | 2 800 | 4 200 | **K95×103×30** | 0.165 |
| | 107 | 21 | 49 000 | 127 000 | 5 000 | 12 900 | 2 700 | 4 000 | **KV100×107×21S** | 0.120 |
| **100** | 107 | 31 $^{-0.3}_{-0.65}$ | 71 500 | 207 000 | 7 300 | 21 100 | 2 700 | 4 000 | **K100×107×31** | 0.173 |
| | 108 | 27 | 61 000 | 153 000 | 6 250 | 15 600 | 2 700 | 4 000 | **K100×108×27** | 0.176 |
| | 108 | 30 | 76 000 | 201 000 | 7 700 | 20 500 | 2 700 | 4 000 | **K100×108×30** | 0.190 |
| | 112 | 21 | 48 500 | 127 000 | 4 950 | 12 900 | 2 500 | 3 800 | **K105×112×21** | 0.130 |
| **105** | 112 | 31 $^{-0.3}_{-0.65}$ | 71 000 | 207 000 | 7 250 | 21 100 | 2 500 | 3 800 | **K105×112×31** | 0.176 |
| | 113 | 30 | 77 500 | 210 000 | 7 900 | 21 400 | 2 500 | 3 800 | **K105×113×30** | 0.198 |
| | 117 | 24 | 54 500 | 149 000 | 5 550 | 15 200 | 2 400 | 3 600 | **K110×117×24** | 0.145 |
| **110** | 117 | 34 $^{-0.3}_{-0.65}$ | 77 500 | 235 000 | 7 900 | 24 000 | 2 400 | 3 600 | **K110×117×34** | 0.205 |
| | 118 | 30 | 79 000 | 219 000 | 8 050 | 22 300 | 2 400 | 3 600 | **K110×118×30** | 0.217 |
| **115** | 123 | 27 $^{-0.3}_{-0.65}$ | 64 000 | 170 000 | 6 550 | 17 300 | 2 300 | 3 500 | **K115×123×27** | 0.200 |
| | 125 | 34 | 95 000 | 241 000 | 9 700 | 24 600 | 2 300 | 3 500 | **K115×125×34** | 0.330 |
| **120** | 127 | 24 $^{-0.3}_{-0.65}$ | 57 500 | 165 000 | 5 850 | 16 800 | 2 200 | 3 300 | **K120×127×24** | 0.160 |

Remarks: Even when an order is placed with reference to “Nominal bearing number” listed in this table, the ordered bearings are subject to being delivered with different cage type.

## Type K




Type K

$F_w$ 120～265mm

| Boundary dimensions mm | | | Basic load ratings dynamic N | static N | dynamic kgf | static kgf | Limiting speeds r/min grease | Limiting speeds r/min oil | Bearing numbers | Mass kg (approx.) |
|---|---|---|---|---|---|---|---|---|---|---|
| $F_w$ | $E_w$ | $B_c$ | $C_r$ | $C_{or}$ | $C_r$ | $C_{or}$ | | | | |
| **120** | 127 | 34 $^{-0.3}_{-0.65}$ | 82 000 | 260 000 | 8 350 | 26 600 | 2 200 | 3 300 | **K120×127×34** | 0.235 |
| **125** | 133 | 35 $^{-0.3}_{-0.65}$ | 87 000 | 260 000 | 8 900 | 26 500 | 2 100 | 3 200 | **K125×133×35** | 0.275 |
| | 135 | 34 | 100 000 | 265 000 | 10 200 | 27 000 | 2 100 | 3 200 | **K125×135×34** | 0.350 |
| **130** | 137 | 24 $^{-0.3}_{-0.65}$ | 59 000 | 175 000 | 6 000 | 17 900 | 2 100 | 3 100 | **K130×137×24** | 0.170 |
| | 137 | 34 | 84 500 | 277 000 | 8 600 | 28 300 | 2 100 | 3 100 | **K130×137×34** | 0.240 |
| **135** | 143 | 35 $^{-0.3}_{-0.65}$ | 92 500 | 288 000 | 9 450 | 29 400 | 2 000 | 3 000 | **K135×143×35** | 0.300 |
| | 150 | 38 | 145 000 | 325 000 | 14 800 | 33 500 | 2 000 | 3 000 | **K135×150×38** | 0.590 |
| **145** | 153 | 26 | 72 000 | 214 000 | 7 350 | 21 800 | 1 900 | 2 800 | **K145×153×26** | 0.250 |
| | 153 | 28 $^{-0.3}_{-0.65}$ | 80 500 | 247 000 | 8 200 | 25 200 | 1 900 | 2 800 | **K145×153×28** | 0.252 |
| | 153 | 36 | 100 000 | 325 000 | 10 200 | 33 000 | 1 900 | 2 800 | **K145×153×36** | 0.335 |
| **150** | 160 | 46 $^{-0.3}_{-0.65}$ | 149 000 | 470 000 | 15 200 | 48 000 | 1 800 | 2 700 | **K150×160×46** | 0.550 |
| **155** | 163 | 26 $^{-0.3}_{-0.65}$ | 73 500 | 224 000 | 7 500 | 22 800 | 1 700 | 2 600 | **K155×163×26** | 0.270 |
| | 163 | 36 | 102 000 | 340 000 | 10 400 | 34 500 | 1 700 | 2 600 | **K155×163×36** | 0.355 |
| **160** | 170 | 46 $^{-0.3}_{-0.65}$ | 155 000 | 505 000 | 15 800 | 51 500 | 1 700 | 2 500 | **K160×170×46** | 0.570 |
| **165** | 173 | 26 | 79 000 | 251 000 | 8 050 | 25 600 | 1 600 | 2 400 | **K165×173×26** | 0.290 |
| | 173 | 32 $^{-0.3}_{-0.65}$ | 97 000 | 330 000 | 9 900 | 33 500 | 1 600 | 2 400 | **K165×173×32** | 0.340 |
| | 173 | 36 | 109 000 | 380 000 | 11 100 | 39 000 | 1 600 | 2 400 | **K165×173×36** | 0.375 |
| **170** | 180 | 46 $^{-0.3}_{-0.65}$ | 160 000 | 540 000 | 16 400 | 55 000 | 1 600 | 2 400 | **K170×180×46** | 0.620 |
| **175** | 183 | 32 $^{-0.3}_{-0.65}$ | 101 000 | 350 000 | 10 300 | 35 500 | 1 500 | 2 300 | **K175×183×32** | 0.360 |
| **185** | 195 | 37 $^{-0.3}_{-0.65}$ | 131 000 | 425 000 | 13 300 | 43 500 | 1 500 | 2 200 | **K185×195×37** | 0.560 |
| **195** | 205 | 37 $^{-0.3}_{-0.65}$ | 135 000 | 450 000 | 13 800 | 46 000 | 1 400 | 2 100 | **K195×205×37** | 0.620 |
| **210** | 220 | 42 $^{-0.3}_{-0.65}$ | 156 000 | 560 000 | 15 900 | 57 000 | 1 300 | 1 900 | **K210×220×42** | 0.740 |
| **220** | 230 | 42 $^{-0.3}_{-0.65}$ | 161 000 | 590 000 | 16 400 | 60 000 | 1 200 | 1 800 | **K220×230×42** | 0.790 |
| **240** | 250 | 42 $^{-0.3}_{-0.65}$ | 167 000 | 635 000 | 17 000 | 64 500 | 1 100 | 1 700 | **K240×250×42** | 0.850 |
| **265** | 280 | 50 $^{-0.3}_{-0.65}$ | 256 000 | 850 000 | 26 100 | 87 000 | 1 000 | 1 500 | **K265×280×50** | 1.810 |

Remarks: Even when an order is placed with reference to “Nominal bearing number” listed in this table, the ordered bearings are subject to being delivered with different cage type.

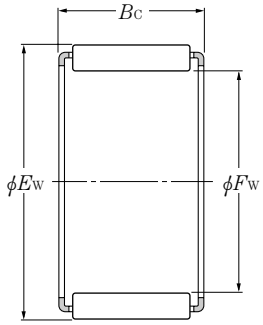
## $F_W$ 285mm

| Boundary dimensions mm | | | Basic load ratings N dynamic | static | kgf dynamic | static | Limiting speeds r/min grease | oil | Bearing numbers | Mass kg |
|---|---|---|---|---|---|---|---|---|---|---|
| $F_w$ | $E_w$ | $B_c$ | $C_r$ | $C_{or}$ | $C_r$ | $C_{or}$ | | | | (approx.) |
| **285** | 300 | 50 $^{-0.3}_{-0.65}$ | 268 000 | 930 000 | 27 300 | 94 500 | 950 | 1 400 | **K285×300×50** | 1.970 |

Remarks: Even when an order is placed with reference to “Nominal bearing number” listed in this table, the ordered bearings are subject to being delivered with different cage type.

## Inch System

### Type PCJ



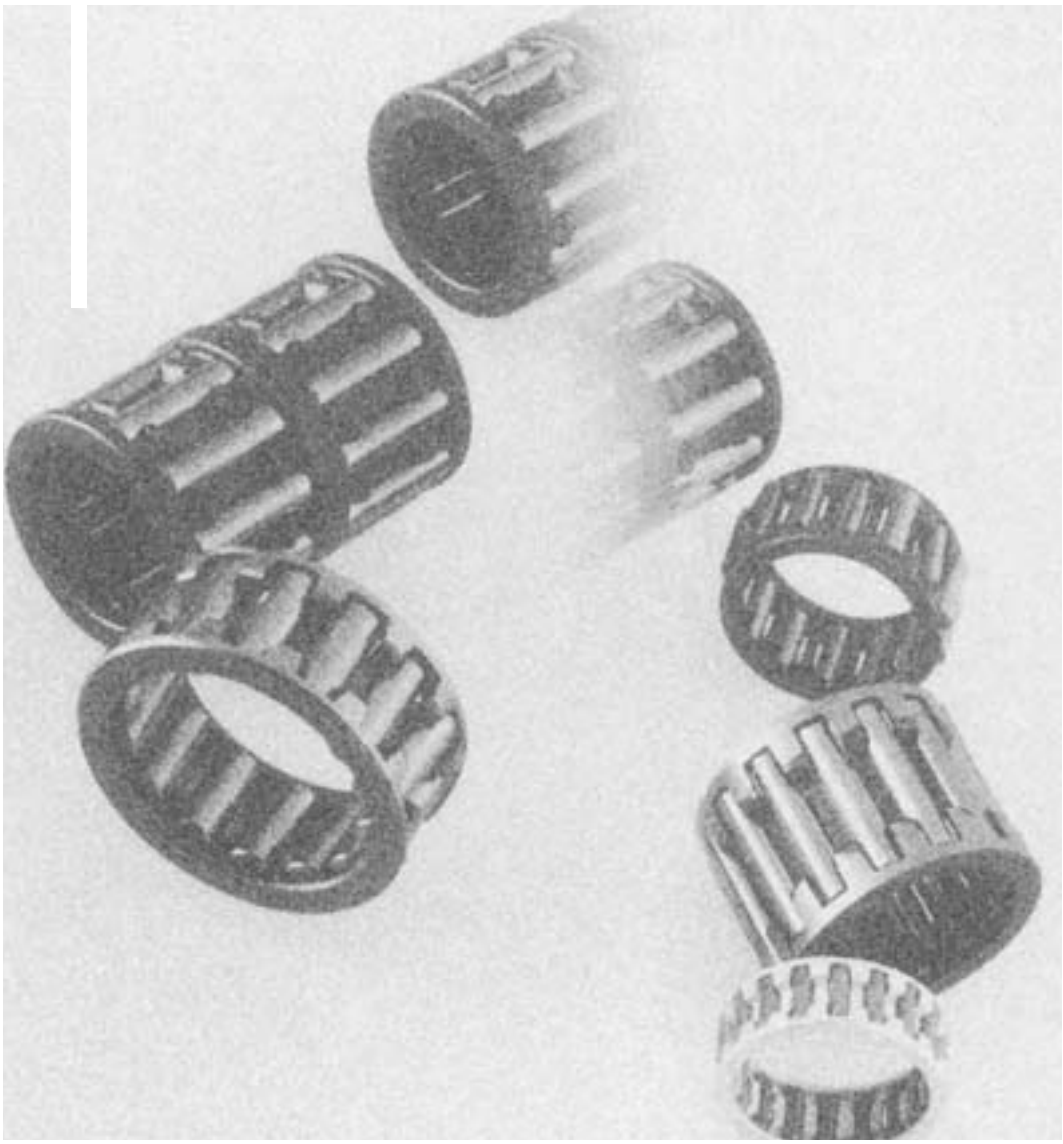

$F_W$ 12.700～63.500mm

| Boundary dimensions mm | | | Basic load ratings dynamic N | static N | dynamic kgf | static kgf | Limiting speeds r/min | | Bearing numbers | Mass kg |
|---|---|---|---|---|---|---|---|---|---|---|
| $F_w$ | $E_w$ | $B_c$ $^{0}_{-0.35}$ ($^{0}_{-0.014}$) | $C_r$ | $C_{or}$ | $C_r$ | $C_{or}$ | grease | oil | | (approx.) |
| **12.700(1/2)** | 15.875(5/8) | 12.70(1/2) | 6 650 | 8 600 | 680 | 875 | 16 000 | 24 000 | **PCJ081008** | 0.003 |
| **19.050(3/4)** | 25.400(1) | 25.40(1) | 25 700 | 35 000 | 2 620 | 3 550 | 13 000 | 20 000 | **PCJ121616** | 0.023 |
| **22.225(7/8)** | 28.575(1 1/8) | 25.40(1) | 29 600 | 44 000 | 3 000 | 4 450 | 11 000 | 17 000 | **PCJ141816** | 0.028 |
| **25.400(1)** | 33.338(1 5/16) | 19.05(3/4) | 29 200 | 38 500 | 2 980 | 3 950 | 10 000 | 15 000 | **PCJ162112** | 0.032 |
| | 33.338(1 5/16) | 25.40(1) | 38 000 | 54 000 | 3 850 | 5 500 | 10 000 | 15 000 | **PCJ162116** | 0.044 |
| **28.575(1 1/8)** | 38.100(1 1/2) | 25.40(1) | 44 000 | 60 000 | 4 500 | 6 100 | 9 500 | 14 000 | **PCJ182416** | 0.058 |
| | 38.100(1 1/2) | 31.75(1 1/4) | 53 500 | 77 000 | 5 450 | 7 850 | 9 500 | 14 000 | **PCJ182420** | 0.073 |
| **31.750(1 1/4)** | 41.275(1 5/8) | 19.05(3/4) | 35 000 | 46 000 | 3 600 | 4 700 | 8 500 | 13 000 | **PCJ202612** | 0.045 |
| | 41.275(1 5/8) | 25.40(1) | 45 500 | 64 500 | 4 650 | 6 600 | 8 500 | 13 000 | **PCJ202616** | 0.062 |
| | 41.275(1 5/8) | 38.10(1 1/2) | 65 000 | 102 000 | 6 650 | 10 400 | 8 500 | 13 000 | **PCJ202624** | 0.096 |
| **34.925(1 3/8)** | 44.450(1 3/4) | 31.75(1 1/4) | 60 000 | 94 500 | 6 150 | 9 650 | 7 500 | 11 000 | **PCJ222820** | 0.090 |
| **38.100(1 1/2)** | 47.625(1 7/8) | 25.40(1) | 51 000 | 78 500 | 5 200 | 8 000 | 6 500 | 10 000 | **PCJ243016** | 0.075 |
| | 47.625(1 7/8) | 31.75(1 1/4) | 62 000 | 101 000 | 6 300 | 10 300 | 6 500 | 10 000 | **PCJ243020** | 0.095 |
| | 47.625(1 7/8) | 38.10(1 1/2) | 72 500 | 124 000 | 7 400 | 12 600 | 6 500 | 10 000 | **PCJ243024** | 0.114 |
| **44.450(1 3/4)** | 53.975(2 1/8) | 25.40(1) | 56 000 | 92 500 | 5 700 | 9 450 | 6 000 | 9 000 | **PCJ283416** | 0.086 |
| | 53.975(2 1/8) | 31.75(1 1/4) | 68 000 | 119 000 | 6 900 | 12 100 | 6 000 | 9 000 | **PCJ283420** | 0.110 |
| | 53.975(2 1/8) | 38.10(1 1/2) | 79 500 | 145 000 | 8 100 | 14 800 | 6 000 | 9 000 | **PCJ283424** | 0.132 |
| **50.800(2)** | 60.325(2 3/8) | 25.40(1) | 58 500 | 102 000 | 5 950 | 10 400 | 5 000 | 7 500 | **PCJ323816** | 0.094 |
| | 60.325(2 3/8) | 31.75(1 1/4) | 71 000 | 131 000 | 7 250 | 13 400 | 5 000 | 7 500 | **PCJ323820** | 0.120 |
| | 60.325(2 3/8) | 38.10(1 1/2) | 83 000 | 160 000 | 8 450 | 16 300 | 5 000 | 7 500 | **PCJ323824** | 0.144 |
| **52.388(2 1/16)** | 61.913(2 7/16) | 25.40(1) | 55 000 | 95 500 | 5 600 | 9 750 | 5 000 | 7 500 | **PCJ333916** | 0.087 |
| **53.975(2 1/8)** | 63.500(2 1/2) | 25.40(1) | 56 500 | 99 500 | 5 750 | 10 200 | 5 000 | 7 500 | **PCJ344016** | 0.091 |
| | 63.500(2 1/2) | 38.10(1 1/2) | 82 000 | 161 000 | 8 350 | 16 400 | 5 000 | 7 500 | **PCJ344024** | 0.144 |
| **57.150(2 1/4)** | 66.675(2 5/8) | 25.40(1) | 57 500 | 104 000 | 5 900 | 10 600 | 4 300 | 6 500 | **PCJ364216** | 0.094 |
| **60.325(2 3/8)** | 69.850(2 3/4) | 38.10(1 1/2) | 87 500 | 182 000 | 8 950 | 18 500 | 4 300 | 6 500 | **PCJ384424** | 0.161 |
| **63.500(2 1/2)** | 73.025(2 7/8) | 25.40(1) | 61 500 | 117 000 | 6 250 | 11 900 | 4 000 | 6 000 | **PCJ404616** | 0.105 |

$F_w$ 63.500～101.600mm

| Boundary dimensions mm | | | Basic load ratings N dynamic | | kgf dynamic | | Limiting speeds r/min | | Bearing numbers | Mass kg |
|---|---|---|---|---|---|---|---|---|---|---|
| $F_w$ | $E_w$ | $B_c$ $^{0}_{-0.35}$ ($^{0}_{-0.014}$) | $C_r$ | static $C_{or}$ | $C_r$ | static $C_{or}$ | grease | oil | | (approx.) |
| **63.500(2½)** | 73.025(2⅞) | 38.10(1½) | 89 000 | 189 000 | 9 100 | 19 300 | 4 000 | 4 000 | **PCJ404624** | 0.167 |
| **69.850(2¾)** | 79.375(3⅛) | 25.40(1) | 65 000 | 130 000 | 6 650 | 13 200 | 3 700 | 5 500 | **PCJ445016** | 0.116 |
| **76.200(3)** | 85.725(3⅜) | 25.40(1) | 68 500 | 143 000 | 7 000 | 14 600 | 3 300 | 5 000 | **PCJ485416** | 0.127 |
| | 85.725(3⅜) | 38.10(1½) | 99 500 | 231 000 | 10 100 | 23 500 | 3 300 | 5 000 | **PCJ485424** | 0.201 |
| **88.900(3½)** | 98.425(3⅞) | 25.40(1) | 75 000 | 169 000 | 7 650 | 17 200 | 2 900 | 4 400 | **PCJ566216** | 0.149 |
| | 101.600(4) | 25.40(1) | 82 000 | 154 000 | 8 350 | 15 700 | 2 900 | 4 400 | **PCJ566416** | 0.187 |
| | 101.600(4) | 38.10(1½) | 121 000 | 253 000 | 12 300 | 25 800 | 2 900 | 4 400 | **PCJ566424** | 0.305 |
| **101.600(4)** | 114.300(4½) | 25.40(1) | 87 500 | 176 000 | 8 950 | 18 000 | 2 500 | 3 800 | **PCJ647216** | 0.212 |
| | 114.300(4½) | 38.10(1½) | 129 000 | 290 000 | 13 200 | 29 500 | 2 500 | 3 800 | **PCJ647224** | 0.345 |

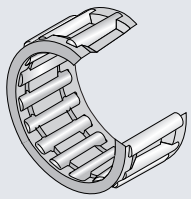
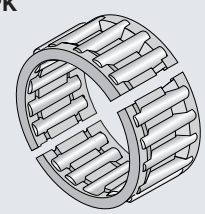
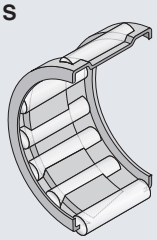
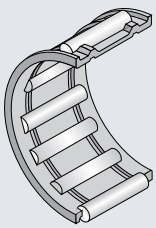
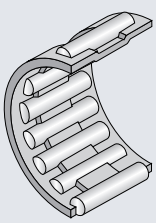
# Needle Roller and Cage Assemblies for connecting rod bearings



# Needle Roller and Cage Assemblies for connecting rod bearings

These needle roller and cage assemblies are specially designed so as to be adaptable to the operating environmental conditions for the connecting rods of small and medium reciprocal engines and compressors.

The connecting rods are used under a severe operating condition wherein acting load magnitude and direction fluctuate rapidly as well as under an high temperature and strict lubrication environment. Therefore, special measures are manly undertaken for the cage structure, material and machining method so that the needle roller and cage assemblies are resistible to the said operating condition and environment.

| Type of needle roller and cage assembly | Location | Cage type | Applied shaft diameter (mm) | Composition of nominal bearing number |
|---|---|---|---|---|
| **PK**  | Large end side | Machined ring type | $\phi$10～$\phi$38 | **PK 20×26×13.8 × 1**<br>Tail code<br>Width<br>Roller set outer diameter<br>Inscribed circle diameter<br>Type code |
| **GPK**  | | | $\phi$10～$\phi$38 | **GPK 20 × 26 × 13.8 X**<br>Tail code<br>Width<br>Roller set outer diameter<br>Inscribed circle diameter<br>Type code |
| **KMJ··S**  | | Welded type | $\phi$10～$\phi$38 | **KMJ 10 × 14 × 8.8 S**<br>Tail code<br>Width<br>Roller set outer diameter<br>Inscribed circle diameter<br>Type code |
| **KBK**  | Small end side | Machined ring type | $\phi$7～$\phi$25 | **KBK 14 × 18 × 17**<br>Width<br>Roller set outer diameter<br>Inscribed circle diameter<br>Type code |
| **KV··S**  | | Welded type | $\phi$7～$\phi$100 | **KV 8 × 11 × 8 S V4**<br>Tail code<br>Tail code<br>Width<br>Roller set outer diameter<br>Inscribed circle diameter<br>Type code |

| Nominal bearing number | Code items and respective dimensions | Remarks |
|---|---|---|
| PK20×26×13.8X1 | Inscribed circle diameter : $\phi$20<br>Roller set outer diamete r: $\phi$26<br>Width : 13.8<br>X1 : numbering up | Cage intended to guide outer ring, eventually surface- treated by non-ferrous plating, etc. |
| GPK20× 26 ×13.8X | Inscribed circle diameter : $\phi$20<br>Roller set outer diameter : $\phi$26<br>Width :13.8 | Cage intended to guide outer ring, eventually surface- treated by non-ferrous plating, etc.<br><br>Can be applied to a crank of integral structure. |
| KMJ10×14×8.8S | Inscribed circle diameter : $\phi$10<br>Roller set outer diameter : $\phi$14<br>Width :8.8<br>S : welded type | Cage intended to guide outer ring, eventually surface- treated by non-ferrous plating, etc. |
| KBK14×18×17 | Inscribed circle diameter : $\phi$14<br>Roller set outer diameter : $\phi$18<br>Width : 17 | Type KBK is intended to guide inner ring, which of the guide surface is designed as long as possible to thereby reduce the surface pressure. In addition, the roller length is so designed as to be maximum against the width of connecting rod, for high load capacity. On the other hand, Type V••S intended for outer ring guide needs a guide along the bore surface of connecting rod. |
| KV8×11×8SV4 | Inscribed circle diameter : $\phi$8<br>Roller set outer diameter : $\phi$11<br>Width : 8<br>S : welded type<br>V4 : numbering up | |

## Radial clearance

**Table 1** shows the recommended clearance values though the radial clearance differs depending on bearing clearance, load, revolutions and ambient temperature.

The proper radial clearance can be got by proper selection and combination of roller diameter, connecting rod hole diameter and pin diameter. **Table 2** shows the examples of selection and combination of those.

**Table 1 Recommended clearance values**

Unit : $\mu$m

| Pin diameter mm Over | incl. | Large end side | Small end side |
|---|---|---|---|
| 6 | 10 | 9～23 | 5～17 |
| 10 | 18 | 10～24 | 5～17 |
| 18 | 30 | 10～24 | 5～17 |
| 30 | 40 | 18～33 | — |

**Table 2 Radial clearance values obtainable by selection and combination**

Case of needle roller and cage assembly Type PK for crank pin
Crank pin hole diameter 22mm H6 (0 to +13 $\mu$m)
Crank pin diameter 14mm h5 (0 to -8 $\mu$m)

Unit : $\mu$m

| Pin diameter sorting class \ Hole diameter sorting class | 0～+4 | +4～+8 | +8～+13 |
|---|---|---|---|
| Sorting class of needle roller used | −4～−6 | −2～−4 | 0～−2 |
| 0～−3 | 10～17 | 10～17 | 10～18 |
| −3～−6 | 13～20 | 13～20 | 13～21 |
| −6～−8 | 16～22 | 16～22 | 16～23 |

## Connecting rod and pin specifications

Connecting rod (bore surface), crank pin and piston pin (outer surface) can be used as direct raceway surface. However, these surfaces must be resistible to great load while maintaining high accuracy. For that, connecting rods and pins acting as the direct raceway surface must comply with the specifications specified hereunder.

Any connecting rods shall be made of cement steel, e.g. chrome molybdenum steel (SCM415, etc.), nickel chrome molybdenum steel (SNCM420, etc.) and any crank pins and piston pins shall also be made of cement steel, e.g. chrome steel (SCr420, etc.), all of which shall be surface-hardened by carburizing.

The surface hardness of each shall range from HRC58 to 64 and adequate depth of effective carburizing-hardened layer shall be secured up to Hv550. The depth of effective carburizing-hardened layer differs depending on actual load and pin diameter.

Feel free to contact **NTN** for the more detailed information.

The respective profile accuracy of connecting rod hole, crank and piston pin outer surfaces shall be as specified in **Table 3**.

Furthermore, the parallelism of crank pin and piston pin shall be 0.02mm or less against 100mm. (**Fig. 1**)

The surface roughness shall be 0.2a for connecting rod and 0.1a for pin outer surface as a guideline.

**Table 3 Recommended accuracy of connecting rod and pin**

Unit : $\mu$m

| Parts | Characteristics | Pin diameter classification mm ～14 | 14～18 | 18～25 | 25～30 | 30～40 |
|---|---|---|---|---|---|---|
| Connecting rod | Roundness (max) | 3 | 4 | 4 | 5 | 5 |
| | Cylindricality (max) | 2 | 3 | 3 | 4 | 4 |
| Pin | Roundness (max) | 2 | 2 | 3 | 3 | 4 |
| | Cylindricality (max) | 1 | 1 | 2 | 2 | 3 |

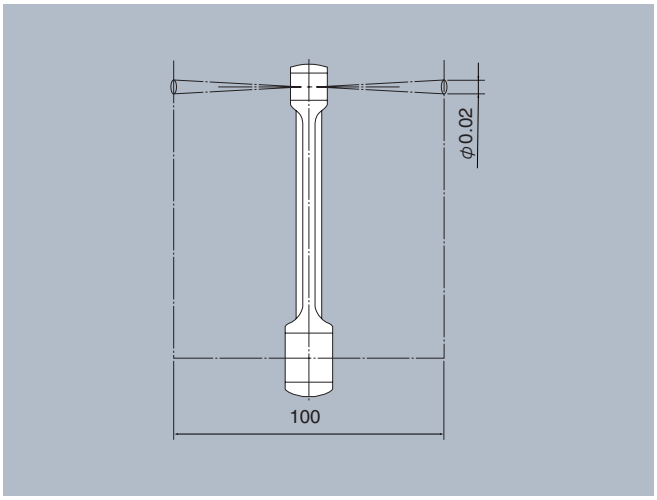



**Fig. 1**

## Regulation to connecting rod

Two methods are available as follows to regulate axial motion of a connecting rod; one method is to make smaller a clearance between the rod and crank web at large end side (Method of regulating at large end side) and another method is to narrow a clearance between the rod and piston boss at the small end side (Method of regulating at small end side).

In general, the method of regulation at large end side is mostly adopted to get accurate motion of a connecting rod.

On the other hand, the method of regulation at small end side is adopted when regulation at large end side disables to secure good lubrication to the bearing and the guide surface at large end side due to short connecting rod and comparatively high revolutions.

### Regulation at large end side

This method regulates axial motion of a connecting rod by forming a sliding surface between the large end of the connecting rod and the side face of crank web. For the use of this method, however, the connecting rod must be provided, on its bore surface, with oil hole, slot and oil groove to feed lubrication oil in and onto the bearing and guide surface at large end side.

The crank web end face shall be surface-hardened as necessary or otherwise a side washer of copper alloy or hardened steel shall be fitted on the guide surface.

On the other hand, the needle roller and cage assembly at small end side shall be guided in axial direction by the side face of piston boss. A great play shall be secured between the piston boss and the connecting rod.

### Regulation at small end side

This method regulates axial motion at the small end face of a connecting rod and the side face of a piston rod. Generally a connecting rod is not provided with oil groove and slot to lubricate the guide surface at small end side, but on occasion it is surface-hardened and uses a side washer.

Usually a connecting rod shall be provided with a through-hole (at its small end side) to lubricate the bearing unit.

On the other hand, at the large end side a free clearance exists between the connecting rod and the crank web and, therefore, generally special measures such as oil groove to lubricate, etc.

may be omitted.

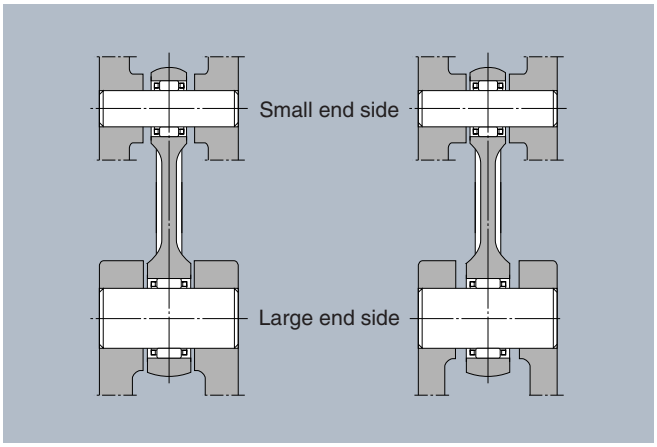
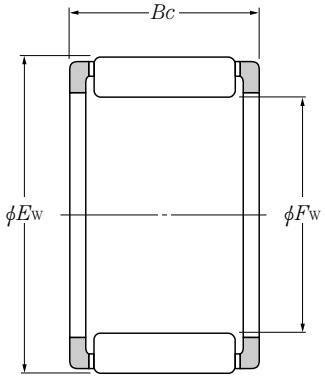



Fig. 2. Regulation at large end side  Fig. 3 Regulation at small end side

## For crank-pin

### Type PK




$F_w$ 10～30mm

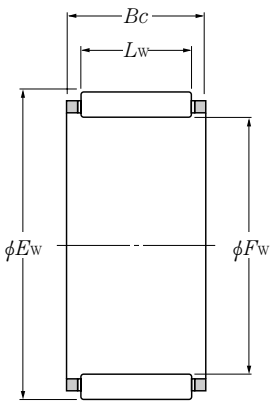
| Boundary dimensions mm | | | Basic load ratings N dynamic | | Basic load ratings kgf dynamic | | Bearing numbers | Mass kg |
|---|---|---|---|---|---|---|---|---|
| $F_w$ | $E_w$ | $B_c$ $^{0}_{-0.2}$ | $C_r$ | $C_{or}$ | $C_r$ | $C_{or}$ | | (approx.) |
| **10** | 14 | 9.8 | 5 050 | 4 900 | 515 | 500 | **PK10×14× 9.8X1** | 0.0037 |
| **12** | 16 | 10 | 5 450 | 5 600 | 555 | 570 | **PK12×16×10.2** | 0.0044 |
| | 17 | 9.8 | 6 800 | 6 550 | 695 | 670 | **PK12×17× 9.8X2** | 0.0051 |
| **14** | 19 | 9.7 | 7 300 | 7 400 | 745 | 755 | **PK14×19× 9.7X1** | 0.0065 |
| **15** | 20 | 9.8 | 7 250 | 7 450 | 740 | 760 | **PK15×20× 9.8X** | 0.0067 |
| | 21 | 11.8 | 10 000 | 10 200 | 1 020 | 1 040 | **PK15×21×11.8X1** | 0.0096 |
| **16** | 22 | 11.8 | 10 000 | 10 300 | 1 020 | 1 050 | **PK16×22×11.8X2** | 0.0097 |
| | 22 | 13.2 | 10 900 | 11 500 | 1 110 | 1 170 | **PK16×22×13.2X** | 0.0110 |
| **18** | 24 | 11.8 | 11 300 | 12 400 | 1 150 | 1 260 | **PK18×24×11.8X3** | 0.0110 |
| | 24 | 13.3 | 13 300 | 15 300 | 1 360 | 1 560 | **PK18×24×13.3X1** | 0.0120 |
| **19** | 24 | 13.9 | 11 900 | 15 200 | 1 220 | 1 550 | **PK19×24×13.9X** | 0.0110 |
| | 25 | 15.8 | 14 300 | 17 000 | 1 460 | 1 730 | **PK19×25×15.8X1** | 0.0150 |
| **20** | 26 | 13.8 | 14 000 | 16 700 | 1 420 | 1 700 | **PK20×26×13.8X6** | 0.0140 |
| **22** | 28 | 15.8 | 15 900 | 20 200 | 1 620 | 2 060 | **PK22×28×15.8X1** | 0.0170 |
| | 29 | 17.8 | 18 800 | 22 800 | 1 920 | 2 320 | **PK22×29×17.8X7** | 0.0240 |
| | 30 | 14.7 | 16 900 | 18 200 | 1 720 | 1 860 | **PK22×30×14.7X2** | 0.0240 |
| | 30 | 17.8 | 21 900 | 25 400 | 2 230 | 2 590 | **PK22×30×17.8X2** | 0.0270 |
| **24** | 31 | 16.8 | 20 800 | 26 600 | 2 120 | 2 710 | **PK24×31×16.8X7** | 0.0240 |
| | 32 | 19.8 | 22 900 | 27 500 | 2 340 | 2 810 | **PK24×32×19.8X6** | 0.0330 |
| **26** | 31 | 13.8 | 14 200 | 20 900 | 1 450 | 2 130 | **PK26×31×13.8X2** | 0.0150 |
| | 34 | 16.8 | 21 600 | 26 100 | 2 200 | 2 660 | **PK26×34×16.8X7** | 0.0320 |
| **27** | 36 | 20.8 | 30 500 | 38 500 | 3 150 | 3 950 | **PK27×36×20.8X1** | 0.0440 |
| **28** | 36 | 15.8 | 23 700 | 30 000 | 2 410 | 3 050 | **PK28×36×15.8X6** | 0.0310 |
| | 37 | 20.8 | 32 500 | 41 500 | 3 300 | 4 250 | **PK28×37×20.8X** | 0.0480 |
| **29** | 39 | 21.4 | 32 500 | 39 500 | 3 300 | 4 000 | **PK29×39×21.4X2** | 0.0550 |
| **30** | 37 | 15.9 | 21 900 | 30 500 | 2 230 | 3 100 | **PK30×37×15.9X** | 0.0280 |
| | 38 | 15.8 | 24 600 | 32 000 | 2 510 | 3 300 | **PK30×38×15.8X** | 0.0320 |

## $F_w$ 30～38mm

| Boundary dimensions mm | | | Basic load ratings N | | Basic load ratings kgf | | Bearing numbers | Mass kg |
|---|---|---|---|---|---|---|---|---|
| $F_w$ | $E_w$ | $B_c$ $^{0}_{-0.2}$ | dynamic $C_r$ | static $C_{or}$ | dynamic $C_r$ | static $C_{or}$ | | (approx.) |
| **30** | 38 | 17.8 | 27 700 | 37 500 | 2 820 | 3 800 | **PK30×38×17.8X1** | 0.037 |
| **31** | 41 | 21.4 | 34 000 | 43 000 | 3 500 | 4 350 | **PK31×41×21.4X** | 0.057 |
| **32** | 43 | 22.4 | 40 000 | 49 500 | 4 100 | 5 050 | **PK32×43×22.4X** | 0.069 |
| **34** | 42 | 19.8 | 31 500 | 45 500 | 3 200 | 4 650 | **PK34×42×19.8X** | 0.046 |
| **38** | 47 | 19.8 | 35 500 | 51 000 | 3 600 | 5 200 | **PK38×47×19.8X1** | 0.056 |

## For piston-pin

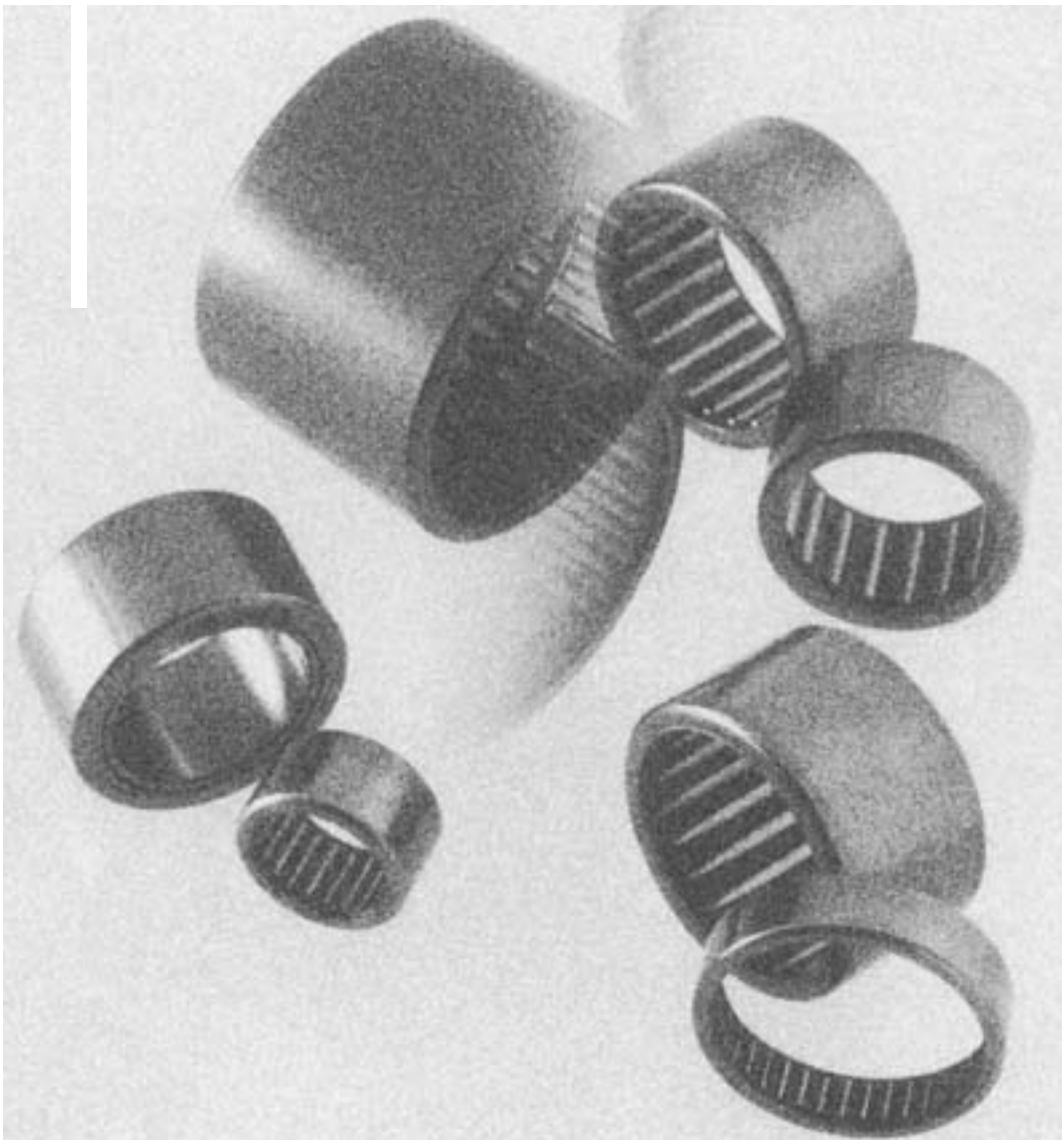
### Type KBK




$F_w$ 7～22mm

| Boundary dimensions mm | | | | Basic load ratings dynamic N | static N | dynamic kgf | static kgf | Bearing numbers | Mass kg |
|---|---|---|---|---|---|---|---|---|---|
| $F_w$ | $E_w$ | $B_c$ $^{0}_{-0.2}$ | $L_w$ | $C_r$ | $C_{or}$ | $C_r$ | $C_{or}$ | | (approx.) |
| **7** | 10 | 9.8 | 6.8 | 3 050 | 2 780 | 310 | 284 | **KBK 7×10× 9.8X** | 0.0022 |
| **8** | 11 | 11.8 | 8.8 | 4 100 | 4 200 | 415 | 430 | **KBK 8×11×11.8X1** | 0.0028 |
| **9** | 12 | 11.5 | 8.8 | 4 400 | 4 750 | 450 | 485 | **KBK 9×12×11.7** | 0.0030 |
| | 14 | 9.8 | 6.8 | 4 500 | 4 200 | 460 | 430 | **KBK10×14× 9.8X** | 0.0042 |
| **10** | 14 | 12.5 | 9.8 | 6 100 | 6 200 | 620 | 635 | **KBK10×14×12.5X1** | 0.0053 |
| | 14 | 14.8 | 9.8 | 6 100 | 6 200 | 620 | 635 | **KBK10×14×14.8X** | 0.0064 |
| **11** | 14 | 13.5 | 10.8 | 5 850 | 7 250 | 595 | 740 | **KBK11×14×13.5X1** | 0.0044 |
| | 15 | 16.4 | 13.8 | 7 500 | 10 200 | 765 | 1 040 | **KBK12×15×16.6** | 0.0055 |
| | 16 | 14.8 | 11.8 | 7 600 | 8 600 | 775 | 875 | **KBK12×16×14.8X** | 0.0072 |
| **12** | 16 | 15.4 | 9.8 | 7 000 | 7 800 | 715 | 795 | **KBK12×16×15.6** | 0.0079 |
| | 16 | 15.8 | 12.8 | 8 100 | 9 350 | 825 | 955 | **KBK12×16×16** | 0.0073 |
| | 17 | 14.8 | 9.8 | 8 400 | 8 550 | 855 | 875 | **KBK12×17×14.8X** | 0.0094 |
| | 18 | 16.8 | 13.8 | 9 750 | 12 400 | 995 | 1 260 | **KBK14×18×17** | 0.0089 |
| **14** | 18 | 19.8 | 13.8 | 9 150 | 11 300 | 930 | 1 160 | **KBK14×18×20** | 0.0130 |
| | 19 | 17.1 | 12.8 | 11 100 | 12 700 | 1 130 | 1 300 | **KBK14×19×17.1X** | 0.0120 |
| **15** | 19 | 17.3 | 13.8 | 10 900 | 14 600 | 1 110 | 1 490 | **KBK15×19×17.3X** | 0.0100 |
| | 20 | 16.8 | 13.8 | 10 800 | 14 700 | 1 100 | 1 500 | **KBK16×20×17** | 0.0100 |
| **16** | 20 | 19.6 | 13.8 | 10 200 | 13 600 | 1 040 | 1 390 | **KBK16×20×19.8** | 0.0130 |
| | 20 | 23.8 | 19.8 | 13 600 | 19 700 | 1 390 | 2 010 | **KBK16×20×23.8X** | 0.0150 |
| | 21 | 19.6 | 15.8 | 13 900 | 17 600 | 1 420 | 1 790 | **KBK16×21×19.6X** | 0.0160 |
| | 21 | 23 | 15.8 | 13 200 | 19 400 | 1 340 | 1 980 | **KBK17×21×23.2** | 0.0160 |
| **17** | 21 | 25 | 17.8 | 13 100 | 19 200 | 1 340 | 1 960 | **KBK17×21×25X** | 0.0170 |
| | 22 | 22 | 18.8 | 16 900 | 22 900 | 1 720 | 2 340 | **KBK17×22×22X1** | 0.0170 |
| **18** | 22 | 21.8 | 15.8 | 12 500 | 18 300 | 1 270 | 1 870 | **KBK18×22×21.8X3** | 0.0150 |
| | 22 | 23.8 | 17.8 | 13 000 | 19 300 | 1 330 | 1 970 | **KBK18×22×23.8X1** | 0.0160 |
| **20** | 25 | 27.9 | 21.8 | 20 800 | 31 500 | 2 120 | 3 200 | **KBK20×25×27.9X** | 0.0270 |
| **22** | 28 | 29.9 | 23.8 | 26 000 | 38 000 | 2 650 | 3 900 | **KBK22×28×29.9X1** | 0.0390 |

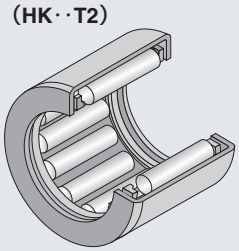
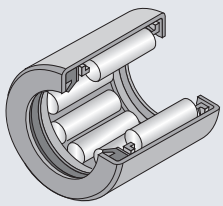
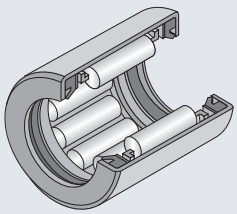
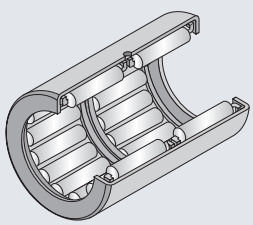
# Drawn Cup Needle Roller Bearings

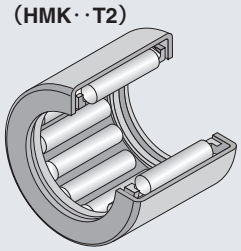
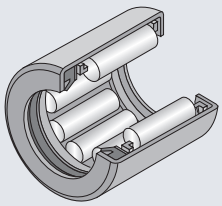
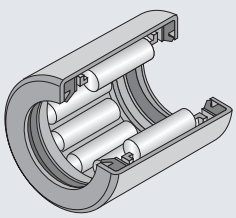
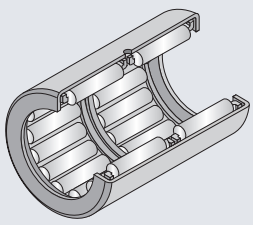


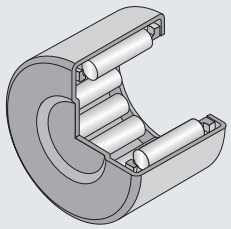
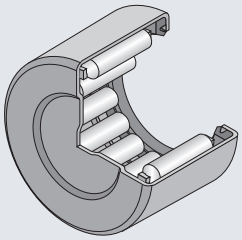
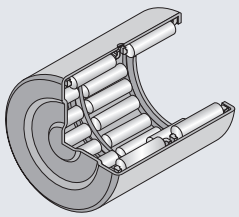
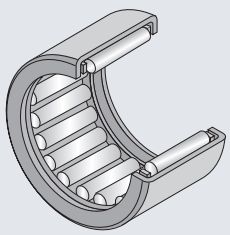
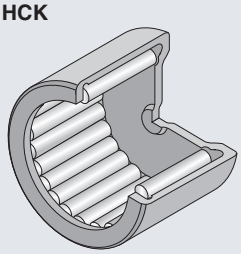
# Drawn Cup Needle Roller Bearings

This bearing type is composed of an outer ring drawn from a thin steel plate by precision drawing, needle rollers and a cage assembled in the outer ring after the raceway surface thereof was hardened. Of the bearings with outer ring, this bearing type is a bearing with the smallest section height which enables space-saving and cost-saving.

Usually design is so made as to use a shaft as the direct raceway surface without using inner ring. The outer ring of this bearing type is of such a construction that the needle rollers and the cage are not separated from one another, so that the bearing is only press-fitted in a rigid housing with proper fit torque. Thus, this bearing type needs no snap ring, etc. to fix itself in axial direction and, in addition, is easy to handle.

| Type of bearing | | | Applied shaft diameter (mm) | Composition of nominal bearing number | Nominal bearing number | Code items and dimensions | Remarks |
|---|---|---|---|---|---|---|---|
| **HK**<br>**(HK··T2)**<br>  | Standard series | Open end | $\phi$3～$\phi$50 | HK 06 09 T2<br>Tail code<br>Width<br>Inscribed circle diameter<br>Type code<br>[Tail code]<br>T2 : Resin cage<br>C : Welding cage | HK0609T2 | Inscribed circle diameter : $\phi$6<br>Width : 9<br>T2 : Resin cage | The bearings with tail code T2 using polyamide resin cage shall be used at allowable temperature 120˚C and, under continuous running, at 100˚C and less. |
| **HK··L**<br>  | | Open end<br>single side seal | $\phi$12～$\phi$50 | HK 20 18 L<br>Tail code<br>Width<br>Inscribed circle diameter<br>Type code | HK2018L | Inscribed circle diameter : $\phi$20<br>Width : 18<br>L : single side seal | This seal type (Tail code : L or LL) synthetic rubber seal built in at its single side or double sides is internally filled with lithium soap base grease.<br>The operating temperature shall range from -25 to 100˚C maximum to prevent deterioration of the seal and filled-up grease.<br><br>The roller length and rated load of this bearing type are shorter and smaller than those of the open type of same dimension. |
| **HK··LL**<br>  | | Open end<br>double-side seal | $\phi$12～$\phi$50 | HK 20 20 LL<br>Tail code<br>Width<br>Inscribed circle diameter<br>Type code | HK2020LL | Inscribed circle diameter : $\phi$20<br>Width : 20<br>LL: Double-side seal | |
| **HK··ZWD**<br>  | | Open end<br>double-row type | $\phi$15～$\phi$30 | HK 20 30 ZW D<br>Tail code<br>Tail code<br>Width<br>Inscribed circle diameter<br>Type code | HK2030ZWD | Inscribed circle diameter : $\phi$20<br>Width : 30<br>ZW : Double-row cage<br>D : Outer ring with oil hole | This type is provided with oil hole on its outer ring. |

| Type of bearing | | | Applied shaft diameter (mm) | Composition of nominal bearing number | Nominal bearing number | Code items and dimensions | Remarks |
|---|---|---|---|---|---|---|---|
| **HMK**<br>**(HMK··T2)**<br>  | Heavy load series | Open end | $\phi$8～$\phi$50 | **HMK 20 15**<br>Width<br>Inscribed circle diameter<br>Type code | HMK2015 | Inscribed circle diameter : $\phi$20<br>Width : 15 | The bearings with tail code T2 using polyamide resin cage shall be used at allowable temperature 120˚C and, under continuous running, at 100˚C and less. |
| **HMK··L**<br>  | | Open end<br>single side seal | $\phi$8～$\phi$50 | **HMK 20 18 L**<br>Tail code<br>Width<br>Inscribed circle diameter<br>Type code | HMK2018L | Inscribed circle diameter : $\phi$20<br>Width : 18<br>L : single side seal | This seal type (Tail code : L or LL) synthetic rubber seal built in at its single side or double sides is internally filled with lithium soap base grease.<br>The operating temperature shall range from -25 to 100˚C maximum to prevent deterioration of the seal and filled-up grease.<br><br>The roller length and rated load of this bearing type are shorter and smaller than those of the open type of same dimension. |
| **HMK··LL**<br>  | | Open end<br>double-side seal | $\phi$8～$\phi$50 | **HMK 20 21 LL**<br>Tail code<br>Width<br>Inscribed circle diameter<br>Type code | HMK2021LL | Inscribed circle diameter : $\phi$20<br>Width : 21<br>LL: Double-side seal | |
| **HMK··ZWD**<br>  | | Open end<br>double-row type | $\phi$38～$\phi$50 | **HMK 38 45 ZW D**<br>Tail code<br>Tail code<br>Width<br>Inscribed circle diameter<br>Type code | HK3845ZWD | Inscribed circle diameter : $\phi$38<br>Width : 45<br>ZW : Double-row cage<br>D : Outer ring with oil hole | This type is provided with oil hole on its outer ring. |

| Type of bearing | | | Applied shaft diameter (mm) | Composition of nominal bearing number | Nominal bearing number | Code items and dimensions | Remarks |
|---|---|---|---|---|---|---|---|
| **BK**<br>**(BK··T2)**<br>  | Standard series | Closed end | $\phi$3～$\phi$50 | **BK 20 20 C**<br>Tail code<br>Width<br>Inscribed circle diameter<br>Type code | BK2020C | Inscribed circle diameter : $\phi$20<br>Width : 20<br>C : Welding cage | The bearings with tail code T2 using polyamide resin cage shall be used at allowable temperature 120˚C and, under continuous running, at 100˚C and less. |
| **BK··L**<br>  | | Closed end<br>single side seal | $\phi$12～$\phi$50 | **BK 20 18 L**<br>Tail code<br>Width<br>Inscribed circle diameter<br>Type code | BK2018L | Inscribed circle diameter : $\phi$20<br>Width : 18<br>L : single side seal | This seal type (Tail code: L) is internally filled up with lithium soap base grease.<br>The operating temperature shall range from -25 to 100˚C maximum to prevent deterioration of the seal and filled-up grease. |
| **BK··ZWD**<br>  | | Closed end<br>double-row type | $\phi$15～$\phi$30 | **BK 20 30 ZW D**<br>Tail code<br>Tail code<br>Width<br>Inscribed circle diameter<br>Type code | BK2030ZWD | Inscribed circle diameter : $\phi$20<br>Width : 30<br>ZW : Double-row cage<br>D : Outer ring with oil hole | Inscribed circle diameter |
| **DCL**<br>  | Inch series | Open end | $\phi$6.35～$\phi$50.8 | **DCL 16 20**<br>Width code<br>Inscribed circle diameter code<br>Type code | DCL1620 | Inscribed circle diameter : $\phi$25.4<br>Width : 31.75 | |
| **HCK**<br>  | Bearing series for universal joints | Closed end | $\phi$10～$\phi$20 | **HCK 16 22**<br>Outer diameter<br>Inscribed circle diameter<br>Type code | HCK1622 | Inscribed circle diameter : $\phi$16<br>Width : $\phi$22 | Full complement roller type with no cage.<br>Already filled up with specific grease. |

## Bearing Fits

It is common that drawn cup needle bearing is press-fitted in a housing by shrinkage fit so post press-fit inscribed circle diameter ($F_w$) comes to ISO Tolerance Rang Class F8. The post press-fit inscribed circle diameter ($F_w$) depends on the housing material and rigidity. It is therefore desirable to decide the interference based on the data measured in pre-testing.

Where the housing rigidity is adequately high, the post press-fit inscribed circle diameter ($F_w$) is secured in nearly F8 range and nearly ordinary radial clearance can be got by adopting the data of bearing fit in housing and on shaft as shown in **Table-1**.

**Table 1 Bearing fit in housing and on shaft**

| Bearing type | Housing | | Shaft | |
|---|---|---|---|---|
| | Iron series | Light alloy | Without inner ring | With inner ring |
| HK,BK<br>HMK,DCL<br>HCK | N6 (N7)<br>J6 (J7)<br>F7 | R6 (R7)<br>M6 (M7)<br>— | h5 (h6)<br>k6 | k5 (j6)<br>— |

## Accuracy of housing bore

Because of its thin-walled outer ring, the performance of drawn cup needle roller bearing is greatly affected by the dimensional accuracy, profile accuracy and bore surface roughness of a housing in which the bearing is press-fitted. To avoid such an inverse affect, the housing bore(hole) must have the accuracy as specified in **Table 2**. For the shaft accuracy refer to **Subsection 7.3 "Accuracy of Shaft and Housing"** (on page A-39).

**Table 2 Accuracy of housing bore**

| Property | Tolerance |
|---|---|
| Roundness (Max) | IT4 or less |
| Cylindricality (Max) | IT4 or less |
| Surface roughness (Max) | 1.6a |

## Oil hole dimension in outer ring

The outer rings of double-row (Tail code : ZW) needle roller and cage assembly Type HK and Type BK are provided with an oil hole to facilitate oil lubrication to the bearing. **Table 3** shows the nominal oil hole diameter.

**Table 3 Diameter of oil hole in outer ring (Metric system)**

| Outer ring diameter | | Nominal oil hole diameter |
|---|---|---|
| Over | incl. | |
| 5 | 10 | 1.5 |
| 10 | 20 | 2.0 |
| 20 | 40 | 2.5 |
| 40 | 80 | 3.0 |
| 80 | 200 | 3.5 |

## Bearing installation

Any drawn cup needle roller bearings must be press-fitted in the housing in specific position by holding a specific jig in perfect contact with the mark-stamped side face of the bearing, without fail.

**Further, hammering directly the bearing ring in installing (press-fitting) is not allowed absolutely.** In installing, it is recommended to use a mandrel with O-ring as illustrated in **Fig.1** as a press-fitting jig. The use of this mandrel would enable to insert easily any drawn cup needle bearing in a housing without risk of twisting and fall-down.

Drawn cup needle roller bearing needs no a snap ring and a shoulder for positioning itself in a housing, **but the bearing must be press-fitted so carefully as not to allow its side face to strike the shoulder for preventing it from deforming, where press-fitted in a housing with shoulder.**

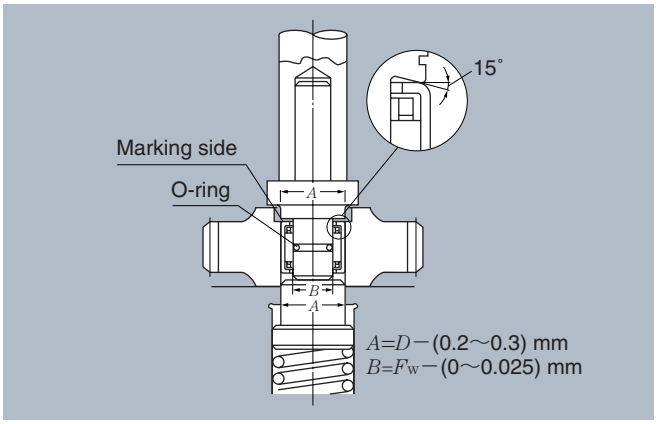



**Fig. 1**

The Type HCK for application to universal joints is fixed to the joint yoke by caulking, using a special-purposed assembler. Feel free to contact **NTN** for any inquiry about the special-purposed assembler (IPH Machine).

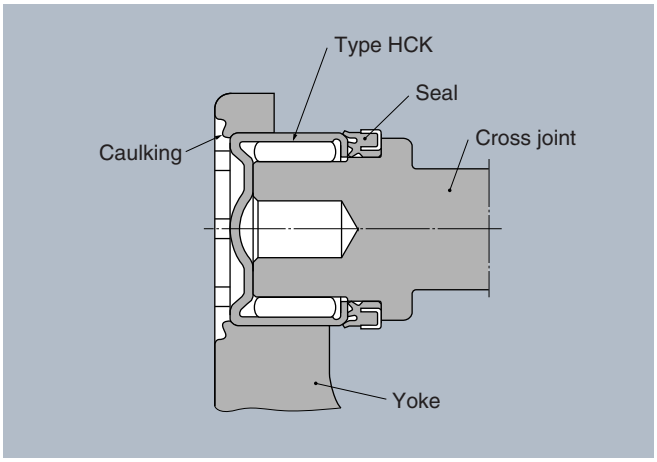



**Fig. 2**

## Bearing Tolerances and Measuring Methods

The outer ring of drawn cup needle roller bearing is so thin-walled that deformation thereof to a certain extent is unavoidable in the manufacturing processes, particularly in the heat-treating process. However, the outer ring is so designed that it is reformed normally from such deformation when being press-fitted in a housing with specific dimensional accuracy and, as a result, it can have the accuracy required to fulfill its specific function.

Hence, **it is meaningless to measure the dimensional accuracy of bearing itself before being press-fitted. So, the following measuring method is used; a bearing to be measured is press-fitted in a linkage of specific dimension (20mm or more in wall thickness) and thereafter the inscribed circle diameter ($F_w$) is measured using a plug gauge or a taper gauge to evaluate the bearing accuracy.**

**Tables 4** to **7** show the dimensional tolerances for the bore diameter of each ring gauge and the inscribed circle diameter ($F_w$) each of standard metric series drawn cup needle roller bearings Type HK and BK, heavy load series Type HMK (metric series), inch series Type DCL, and inch series HCK for application to universal joints. When measuring the inscribed circle diameter of roller, the low tolerance for the inscribed circle diameter shall be applied to the dimension at "GO" side and the value of the high tolerance for the inscribed circle diameter plus (+) 2 $\mu$m shall be applied to the dimension at "NOT GO" side in compliance with the relevant ISO standard.

**Table 4 Dimensional tolerance for inscribed circle diameter (Type HK and BK)**

Unit : mm

| Nominal inscribed circle dia. $F_w$ | Nominal outer ring outer dia. $D$ | Ring gauge bore dia. | Tolerance for inscribed circle diameter | |
|---|---|---|---|---|
| | | | High | Low |
| 3 | 6.5 | 6.484 | 3.016 | 3.006 |
| 4 | 8 | 7.984 | 4.022 | 4.010 |
| 5 | 9 | 8.984 | 5.022 | 5.010 |
| 6 | 10 | 9.984 | 6.022 | 6.010 |
| 7 | 11 | 10.980 | 7.028 | 7.013 |
| 8 | 12 | 11.980 | 8.028 | 8.013 |
| 9 | 13 | 12.980 | 9.028 | 9.013 |
| 10 | 14 | 13.980 | 10.028 | 10.013 |
| 12 | 16 | 15.980 | 12.034 | 12.016 |
| 12 | 18 | 17.980 | 12.034 | 12.016 |
| 13 | 19 | 18.976 | 13.034 | 13.016 |
| 14 | 20 | 19.976 | 14.034 | 14.016 |
| 15 | 21 | 20.976 | 15.034 | 15.016 |
| 16 | 22 | 21.976 | 16.034 | 16.016 |
| 17 | 23 | 22.976 | 17.034 | 17.016 |
| 18 | 24 | 23.976 | 18.034 | 18.016 |
| 20 | 26 | 25.976 | 20.041 | 20.020 |
| 22 | 28 | 27.976 | 22.041 | 22.020 |
| 25 | 32 | 31.972 | 25.041 | 25.020 |
| 28 | 35 | 34.972 | 28.041 | 28.020 |
| 30 | 37 | 36.972 | 30.041 | 30.020 |
| 35 | 42 | 41.972 | 35.050 | 35.025 |
| 40 | 47 | 46.972 | 40.050 | 40.025 |
| 45 | 52 | 51.967 | 45.050 | 45.025 |
| 50 | 58 | 57.967 | 50.050 | 50.025 |
| 55 | 63 | 62.967 | 55.060 | 55.030 |
| 60 | 68 | 67.967 | 60.060 | 60.030 |

**Table 5 Dimensional tolerance for inscribed circle diameter (Type HMK)**

Unit : mm

| Nominal inscribed circle dia. $F_w$ | Nominal outer ring outer dia. $D$ | Ring gauge bore dia. | Tolerance for inscribed circle diameter | |
|---|---|---|---|---|
| | | | High | Low |
| 8 | 15 | 14.995 | 8.028 | 8.013 |
| 9 | 16 | 15.995 | 9.028 | 9.013 |
| 10 | 17 | 16.995 | 10.028 | 10.013 |
| 12 | 19 | 18.995 | 12.034 | 12.016 |
| 14 | 22 | 21.995 | 14.034 | 14.016 |
| 15 | 22 | 21.995 | 15.034 | 15.016 |
| 16 | 24 | 23.995 | 16.034 | 16.016 |
| 17 | 24 | 23.995 | 17.034 | 17.016 |
| 18 | 25 | 24.995 | 18.034 | 18.016 |
| 19 | 27 | 26.995 | 19.041 | 19.020 |
| 20 | 27 | 26.995 | 20.041 | 20.020 |
| 21 | 29 | 28.995 | 21.041 | 21.020 |
| 22 | 29 | 28.995 | 22.041 | 22.020 |
| 24 | 31 | 30.994 | 24.041 | 24.020 |
| 25 | 33 | 32.994 | 25.041 | 25.020 |
| 26 | 34 | 33.994 | 26.041 | 26.020 |
| 28 | 37 | 36.994 | 28.041 | 28.020 |
| 29 | 38 | 37.994 | 29.041 | 29.020 |
| 30 | 40 | 39.994 | 30.041 | 30.020 |
| 32 | 42 | 41.994 | 32.050 | 32.025 |
| 35 | 45 | 44.994 | 35.050 | 35.025 |
| 37 | 47 | 46.994 | 37.050 | 37.025 |
| 38 | 48 | 47.994 | 38.050 | 38.025 |
| 40 | 50 | 49.994 | 40.050 | 40.025 |
| 45 | 55 | 54.994 | 45.050 | 45.025 |
| 50 | 62 | 61.994 | 50.050 | 50.025 |

**Table 6 Dimensional tolerance for inscribed circle diameter (Type DCL)**

Unit : mm

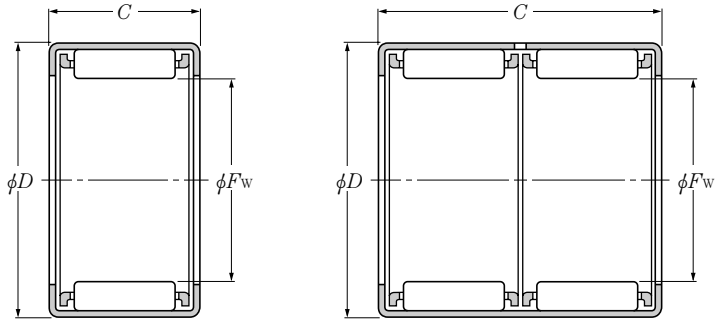
| Nominal inscribed circle dia. $F_w$ | Nominal outer ring outer dia. $D$ | Ring gauge bore dia. | Tolerance for inscribed circle diameter | |
|---|---|---|---|---|
| | | | High | Low |
| 6.350 | 11.112 | 11.125 | 6.411 | 6.388 |
| 7.938 | 12.700 | 12.713 | 7.998 | 7.976 |
| 9.525 | 14.288 | 14.300 | 9.586 | 9.563 |
| 11.112 | 15.875 | 15.888 | 11.173 | 11.151 |
| 12.700 | 17.462 | 17.475 | 12.761 | 12.738 |
| 14.288 | 19.050 | 19.063 | 14.348 | 14.326 |
| 15.875 | 20.638 | 20.650 | 15.936 | 15.913 |
| 17.462 | 22.225 | 22.238 | 17.523 | 17.501 |
| 19.050 | 25.400 | 25.387 | 19.086 | 19.063 |
| 20.638 | 26.988 | 26.975 | 20.673 | 20.650 |
| 22.225 | 28.575 | 28.562 | 22.261 | 22.238 |
| 23.812 | 30.162 | 30.150 | 23.848 | 23.825 |
| 25.400 | 31.750 | 31.737 | 25.436 | 25.413 |
| 26.988 | 33.338 | 33.325 | 27.023 | 27.000 |
| 28.575 | 34.925 | 34.912 | 28.611 | 28.588 |
| 30.162 | 38.100 | 38.087 | 30.198 | 30.175 |
| 31.750 | 38.100 | 38.087 | 31.786 | 31.763 |
| 34.925 | 41.275 | 41.262 | 34.963 | 34.938 |
| 38.100 | 47.625 | 47.612 | 38.141 | 38.113 |
| 41.275 | 50.800 | 50.787 | 41.316 | 41.288 |
| 44.450 | 53.975 | 53.962 | 44.493 | 44.463 |
| 47.625 | 57.150 | 57.137 | 47.668 | 47.638 |
| 50.800 | 60.325 | 60.312 | 50.846 | 50.815 |

**Table 7 Dimensional tolerance for inscribed circle diameter (Type HCK)**

Unit : mm

| Nominal inscribed circle dia. $F_w$ | Nominal outer ring outer dia. $D$ | Ring gauge bore dia. | Tolerance for inscribed circle diameter | |
|---|---|---|---|---|
| | | | High | Low |
| 10 | 15 | 15.016 | 10.026 | 10.011 |
| 11.656 | 17.1 | 17.116 | 11.687 | 11.669 |
| 13 | 19 | 19.020 | 13.031 | 13.013 |
| 14 | 20 | 20.020 | 14.031 | 14.013 |
| 16 | 22 | 22.020 | 16.031 | 16.013 |
| 18 | 24 | 24.020 | 18.031 | 18.013 |
| 18 | 24.6 | 24.620 | 18.031 | 18.013 |
| 20 | 27.9 | 27.920 | 20.038 | 20.017 |

## Type HK, Type HK··ZWD
## Type HMK
## Type BK,,Type BK··ZWD



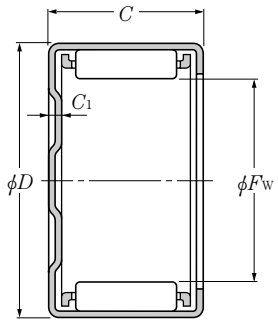

Type HK, Type HMK　　Type HK··ZWD

**Opened End Type**

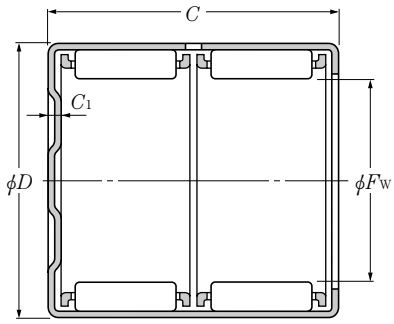
$F_w$ 3～10mm

| Boundary dimensions mm | | | | Basic load ratings dynamic N | static N | dynamic kgf | static kgf | Limiting speeds r/min grease | oil | Bearing numbers open end design | closed end design | Mass kg (approx.) | Appropriate[1] inner ring (as a reference) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $F_w$ | $D$ | $C$ $^{0}_{-0.2}$ | $C_1$ max | $C_r$ | $C_{or}$ | $C_r$ | $C_{or}$ | | | | | | |
| 3 | 6.5 | 6 | — | 925 | 565 | 94 | 58 | 33 000 | 50 000 | **HK0306T2** | — | 0.0006 | — |
| | 6.5 | 6 | 0.8 | 925 | 565 | 94 | 58 | 33 000 | 50 000 | — | **BK0306T2** | 0.0007 | — |
| 4 | 8 | 8 | — | 1 770 | 1 270 | 180 | 129 | 30 000 | 45 000 | **HK0408T2** | — | 0.0016 | — |
| | 8 | 8 | 1.6 | 1 770 | 1 270 | 180 | 129 | 30 000 | 45 000 | — | **BK0408T2** | 0.0018 | — |
| 5 | 9 | 9 | — | 2 640 | 2 190 | 269 | 224 | 27 000 | 40 000 | **HK0509T2** | — | 0.0019 | — |
| | 9 | 9 | 1.6 | 2 640 | 2 190 | 269 | 224 | 27 000 | 40 000 | — | **BK0509T2** | 0.0021 | — |
| 6 | 10 | 9 | — | 2 660 | 2 280 | 272 | 233 | 25 000 | 37 000 | **HK0609T2** | — | 0.0022 | — |
| | 10 | 9 | 1.6 | 2 660 | 2 280 | 272 | 233 | 25 000 | 37 000 | — | **BK0609T2** | 0.0024 | — |
| 7 | 11 | 9 | — | 3 400 | 3 250 | 345 | 330 | 23 000 | 34 000 | **HK0709** | — | 0.0025 | — |
| | 11 | 9 | 1.6 | 3 400 | 3 250 | 345 | 330 | 23 000 | 34 000 | — | **BK0709** | 0.0027 | — |
| 8 | 12 | 10 | — | 3 850 | 3 950 | 395 | 400 | 20 000 | 30 000 | **HK0810C** | — | 0.0032 | IR 5× 8×12 |
| | 12 | 10 | 1.6 | 3 850 | 3 950 | 395 | 400 | 20 000 | 30 000 | — | **BK0810C** | 0.0034 | IR 5× 8×12 |
| | 15 | 10 | — | 3 800 | 2 870 | 390 | 293 | 20 000 | 30 000 | **HMK0810** | — | 0.0067 | IR 5× 8×12 |
| | 15 | 15 | — | 6 600 | 5 800 | 675 | 590 | 20 000 | 30 000 | **HMK0815** | — | 0.0100 | IR 5× 8×16 |
| | 15 | 20 | — | 9 050 | 8 750 | 925 | 890 | 20 000 | 30 000 | **HMK0820T2** | — | 0.0130 | — |
| 9 | 13 | 10 | — | 4 750 | 5 300 | 485 | 540 | 18 000 | 27 000 | **HK0910** | — | 0.0035 | IR 6× 9×12 |
| | 13 | 10 | 1.6 | 4 750 | 5 300 | 485 | 540 | 18 000 | 27 000 | — | **BK0910** | 0.0039 | IR 6× 9×12 |
| | 13 | 12 | — | 5 650 | 6 650 | 575 | 680 | 18 000 | 27 000 | **HK0912** | — | 0.0042 | IR 6× 9×12 |
| | 13 | 12 | 1.6 | 5 650 | 6 650 | 575 | 680 | 18 000 | 27 000 | — | **BK0912** | 0.0045 | IR 6× 9×12 |
| | 16 | 12 | — | 5 300 | 4 450 | 540 | 455 | 18 000 | 27 000 | **HMK0912** | — | 0.0087 | IR 6× 9×16 |
| | 16 | 16 | — | 7 400 | 6 850 | 755 | 700 | 18 000 | 27 000 | **HMK0916** | — | 0.0120 | — |
| 10 | 14 | 10 | — | 4 500 | 5 100 | 460 | 520 | 16 000 | 24 000 | **HK1010** | — | 0.0038 | IR 7×10×10.5 |
| | 14 | 10 | 1.6 | 4 500 | 5 100 | 460 | 520 | 16 000 | 24 000 | — | **BK1010** | 0.0042 | IR 7×10×10.5 |
| | 14 | 12 | — | 5 900 | 7 250 | 605 | 735 | 16 000 | 24 000 | **HK1012** | — | 0.0045 | IR 7×10×16 |
| | 14 | 12 | 1.6 | 5 900 | 7 250 | 605 | 735 | 16 000 | 24 000 | — | **BK1012** | 0.0050 | IR 7×10×16 |
| | 14 | 15 | — | 7 100 | 9 150 | 725 | 935 | 16 000 | 24 000 | **HK1015** | — | 0.0056 | IR 7×10×16 |
| | 14 | 15 | 1.6 | 7 100 | 9 150 | 725 | 935 | 16 000 | 24 000 | — | **BK1015** | 0.0062 | IR 7×10×16 |
| | 17 | 10 | — | 4 250 | 3 450 | 435 | 350 | 16 000 | 24 000 | **HMK1010** | — | 0.0079 | IR 7×10×10.5 |
| | 17 | 12 | — | 5 600 | 4 850 | 570 | 495 | 16 000 | 24 000 | **HMK1012** | — | 0.0094 | IR 7×10×16 |

Note 1) Bearing with inner ring is represented by HK+IR. (Refer to "Inner Ring Dimensions Table" on page B-117.)
EX. HK1012 + IR7×10×16.5




Type BK




Type BK··ZWD

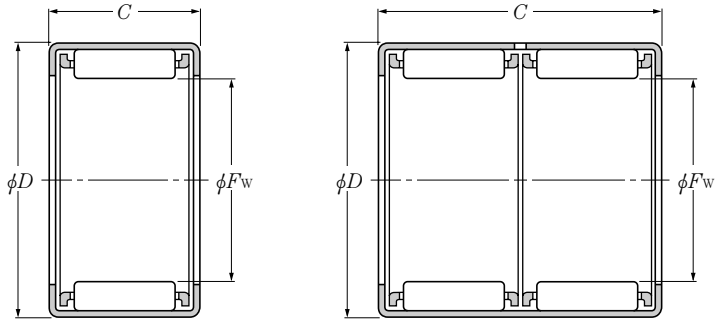
Closed End Type

$F_w$ 10～16mm

| Boundary dimensions mm | | | | Basic load ratings dynamic N | static N | dynamic kgf | static kgf | Limiting speeds r/min grease | oil | Bearing numbers open end design | closed end design | Mass kg (approx.) | Appropriate[1] inner ring (as a reference) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $F_w$ | $D$ | $C$ $^{0}_{-0.2}$ | $C_1$ max | $C_r$ | $C_{or}$ | $C_r$ | $C_{or}$ | | | | | | |
| 10 | 17 | 15 | — | 7 400 | 6 950 | 755 | 710 | 16 000 | 24 000 | **HMK1015** | — | 0.0120 | IR 7×10×16 |
| | 17 | 20 | — | 10 200 | 10 500 | 1 040 | 1 070 | 16 000 | 24 000 | **HMK1020** | — | 0.0160 | — |
| 12 | 16 | 10 | — | 5 050 | 6 250 | 515 | 635 | 13 000 | 20 000 | **HK1210** | — | 0.0046 | IR 8×12×10.5 |
| | 16 | 10 | 1.6 | 5 050 | 6 250 | 515 | 635 | 13 000 | 20 000 | — | **BK1210** | 0.0052 | IR 8×12×10.5 |
| | 18 | 12 | — | 6 600 | 7 300 | 675 | 745 | 13 000 | 20 000 | **HK1212** | — | 0.0091 | IR 8×12×12.5 |
| | 18 | 12 | 2.7 | 6 600 | 7 300 | 675 | 745 | 13 000 | 20 000 | — | **BK1212** | 0.0100 | IR 8×12×12.5 |
| | 19 | 12 | — | 7 100 | 6 900 | 725 | 705 | 13 000 | 20 000 | **HMK1212** | — | 0.0110 | IR 8×12×12.5 |
| | 19 | 15 | — | 9 400 | 9 900 | 955 | 1 010 | 13 000 | 20 000 | **HMK1215** | — | 0.0140 | IR 9×12×16 |
| | 19 | 20 | — | 12 300 | 14 000 | 1 260 | 1 430 | 13 000 | 20 000 | **HMK1220** | — | 0.0180 | — |
| | 19 | 25 | — | 15 300 | 18 600 | 1 560 | 1 890 | 13 000 | 20 000 | **HMK1225** | — | 0.0230 | — |
| 13 | 19 | 12 | — | 6 950 | 7 900 | 705 | 805 | 12 000 | 18 000 | **HK1312** | — | 0.0100 | IR10×13×12.5 |
| | 19 | 12 | 2.7 | 6 950 | 7 900 | 705 | 805 | 12 000 | 18 000 | — | **BK1312** | 0.0110 | IR10×13×12.5 |
| 14 | 20 | 12 | — | 7 200 | 8 500 | 735 | 865 | 11 000 | 17 000 | **HK1412** | — | 0.0110 | IR10×14×13 |
| | 20 | 12 | 2.7 | 7 200 | 8 500 | 735 | 865 | 11 000 | 17 000 | — | **BK1412** | 0.0120 | IR10×14×13 |
| | 20 | 16 | — | 10 700 | 14 000 | 1 090 | 1 430 | 11 000 | 17 000 | **HK1416** | — | 0.0150 | — |
| | 20 | 16 | 2.7 | 10 700 | 14 000 | 1 090 | 1 430 | 11 000 | 17 000 | — | **BK1416** | 0.0160 | — |
| | 22 | 16 | — | 11 500 | 12 000 | 1 180 | 1 220 | 11 000 | 17 000 | **HMK1416** | — | 0.0190 | IR10×14×20 |
| | 22 | 20 | — | 14 600 | 16 200 | 1 490 | 1 650 | 11 000 | 17 000 | **HMK1420** | — | 0.0240 | — |
| 15 | 21 | 12 | — | 7 500 | 9 100 | 765 | 930 | 11 000 | 16 000 | **HK1512** | — | 0.0110 | IR12×15×12.5 |
| | 21 | 12 | 2.7 | 7 500 | 9 100 | 765 | 930 | 11 000 | 16 000 | — | **BK1512** | 0.0130 | IR12×15×12.5 |
| | 21 | 16 | — | 10 700 | 14 400 | 1 090 | 1 470 | 11 000 | 16 000 | **HK1516** | — | 0.0150 | IR12×15×16.5 |
| | 21 | 16 | 2.7 | 10 700 | 14 400 | 1 090 | 1 470 | 11 000 | 16 000 | — | **BK1516** | 0.0170 | IR12×15×16.5 |
| | 21 | 22 | — | 12 900 | 18 200 | 1 310 | 1 860 | 11 000 | 16 000 | **HK1522ZWD** | — | 0.0200 | IR12×15×22.5 |
| | 21 | 22 | 2.7 | 12 900 | 18 200 | 1 310 | 1 860 | 11 000 | 16 000 | — | **BK1522ZWD** | 0.0220 | IR12×15×22.5 |
| | 22 | 10 | — | 6 100 | 6 000 | 620 | 610 | 11 000 | 16 000 | **HMK1510** | — | 0.0110 | IR10×15×12.5 |
| | 22 | 12 | — | 7 950 | 8 450 | 810 | 860 | 11 000 | 16 000 | **HMK1512** | — | 0.0130 | IR12×15×12.5 |
| | 22 | 15 | — | 10 900 | 12 700 | 1 120 | 1 300 | 11 000 | 16 000 | **HMK1515** | — | 0.0160 | IR12×15×16 |
| | 22 | 20 | — | 14 900 | 18 900 | 1 510 | 1 920 | 11 000 | 16 000 | **HMK1520** | — | 0.0220 | IR12×15×22.5 |
| | 22 | 25 | — | 18 500 | 25 000 | 1 880 | 2 550 | 11 000 | 16 000 | **HMK1525** | — | 0.0270 | — |
| 16 | 22 | 12 | — | 7 750 | 9 700 | 795 | 990 | 10 000 | 15 000 | **HK1612** | — | 0.0120 | IR12×16×13 |

Note 1) Bearing with inner ring is represented by HK+IR. (Refer to "Inner Ring Dimensions Table" on page B-117, B118.)
EX. HK1312 + IR10×13×12.5

## Type HK, Type HK··ZWD
## Type HMK
## Type BK, Type BK··ZWD




Type HK, Type HMK　　Type HK··ZWD

**Opened End Type**

$F_w$ 16～20mm

| Boundary dimensions mm | | | | Basic load ratings dynamic N | static N | dynamic kgf | static kgf | Limiting speeds r/min grease | oil | Bearing numbers open end design | closed end design | Mass kg (approx.) | Appropriate[1] inner ring (as a reference) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $F_w$ | $D$ | $C$ $^{0}_{-0.2}$ | $C_1$ max | $C_r$ | $C_{or}$ | $C_r$ | $C_{or}$ | | | | | | |
| | 22 | 12 | 2.7 | 7 750 | 9 700 | 795 | 990 | 10 000 | 15 000 | — | **BK1612** | 0.014 | IR12×16×13 |
| | 22 | 16 | — | 11 100 | 15 300 | 1 130 | 1 560 | 10 000 | 15 000 | **HK1616** | — | 0.016 | IR12×16×20 |
| | 22 | 16 | 2.7 | 11 100 | 15 300 | 1 130 | 1 560 | 10 000 | 15 000 | — | **BK1616** | 0.018 | IR12×16×20 |
| **16** | 22 | 22 | — | 13 300 | 19 400 | 1 360 | 1 980 | 10 000 | 15 000 | **HK1622ZWD** | — | 0.022 | — |
| | 22 | 22 | 2.7 | 13 300 | 19 400 | 1 360 | 1 980 | 10 000 | 15 000 | — | **BK1622ZWD** | 0.023 | — |
| | 24 | 16 | — | 12 400 | 13 500 | 1 260 | 1 370 | 10 000 | 15 000 | **HMK1616** | — | 0.021 | IR12×16×20 |
| | 24 | 20 | — | 15 600 | 18 200 | 1 590 | 1 860 | 10 000 | 15 000 | **HMK1620** | — | 0.027 | IR12×16×22 |
| | 23 | 12 | — | 8 500 | 11 100 | 865 | 1 130 | 9 500 | 14 000 | **HK1712** | — | 0.012 | — |
| | 23 | 12 | 2.7 | 8 500 | 11 100 | 865 | 1 130 | 9 500 | 14 000 | — | **BK1712** | 0.015 | — |
| **17** | 24 | 15 | — | 12 100 | 15 000 | 1 230 | 1 530 | 9 500 | 14 000 | **HMK1715** | — | 0.018 | IR14×17×17 |
| | 24 | 20 | — | 15 200 | 20 000 | 1 540 | 2 040 | 9 500 | 14 000 | **HMK1720C** | — | 0.024 | IR12×17×20.5 |
| | 24 | 25 | — | 19 300 | 27 400 | 1 970 | 2 790 | 9 500 | 14 000 | **HMK1725B** | — | 0.030 | IR12×17×25.5 |
| | 24 | 12 | — | 8 300 | 10 900 | 845 | 1 110 | 8 500 | 13 000 | **HK1812** | — | 0.013 | IR15×18×12.5 |
| | 24 | 12 | 2.7 | 8 300 | 10 900 | 845 | 1 110 | 8 500 | 13 000 | — | **BK1812** | 0.015 | IR15×18×12.5 |
| | 24 | 16 | — | 11 800 | 17 300 | 1 210 | 1 760 | 8 500 | 13 000 | **HK1816** | — | 0.018 | IR15×18×16.5 |
| | 24 | 16 | 2.7 | 11 800 | 17 300 | 1 210 | 1 760 | 8 500 | 13 000 | — | **BK1816** | 0.020 | IR15×18×16.5 |
| **18** | 25 | 13 | — | 10 200 | 12 200 | 1 040 | 1 240 | 8 500 | 13 000 | **HMK1813** | — | 0.016 | IR15×18×16 |
| | 25 | 15 | — | 12 000 | 15 100 | 1 220 | 1 540 | 8 500 | 13 000 | **HMK1815** | — | 0.019 | IR15×18×16 |
| | 25 | 17 | — | 13 800 | 18 000 | 1 400 | 1 830 | 8 500 | 13 000 | **HMK1817** | — | 0.021 | IR15×18×17.5 |
| | 25 | 19 | — | 15 500 | 20 900 | 1 580 | 2 130 | 8 500 | 13 000 | **HMK1819** | — | 0.024 | IR15×18×20.5 |
| | 25 | 20 | — | 16 300 | 22 300 | 1 660 | 2 280 | 8 500 | 13 000 | **HMK1820** | — | 0.025 | IR15×18×20.5 |
| | 25 | 25 | — | 20 300 | 29 600 | 2 070 | 3 000 | 8 500 | 13 000 | **HMK1825** | — | 0.031 | IR15×18×25.5 |
| **19** | 27 | 16 | — | 13 900 | 16 300 | 1 410 | 1 660 | 8 500 | 13 000 | **HMK1916** | — | 0.025 | IR15×19×20 |
| | 27 | 20 | — | 17 500 | 22 100 | 1 790 | 2 250 | 8 500 | 13 000 | **HMK1920** | — | 0.031 | — |
| | 26 | 12 | — | 9 250 | 13 000 | 945 | 1330 | 8 000 | 12 000 | **HK2012C** | — | 0.014 | IR15×20×13 |
| | 26 | 12 | 2.7 | 9 250 | 13 000 | 945 | 1330 | 8 000 | 12 000 | — | **BK2012** | 0.017 | IR15×20×13 |
| **20** | 26 | 16 | — | 13 000 | 20 100 | 1 320 | 2 050 | 8 000 | 12 000 | **HK2016** | — | 0.019 | IR17×20×16.5 |
| | 26 | 16 | 2.7 | 13 000 | 20 100 | 1 320 | 2 050 | 8 000 | 12 000 | — | **BK2016** | 0.022 | IR17×20×16.5 |
| | 26 | 20 | — | 16 400 | 27 100 | 1 670 | 2 760 | 8 000 | 12 000 | **HK2020C** | — | 0.024 | IR17×20×20.5 |
| | 26 | 20 | 2.7 | 16 400 | 27 100 | 1 670 | 2 760 | 8 000 | 12 000 | — | **BK2020C** | 0.027 | IR17×20×20.5 |

Note 1) Bearing with inner ring is represented by HK+IR. (Refer to "Inner Ring Dimensions Table" on page B-118, B119.)
EX. HK1812 + IR15×18×12.5